

Callum Mackenzie

403-473-1818 | callum@camackenzie.com | camackenzie.com | linkedin.com/in/callum-ma | github.com/CallumMackenzie

EXPERIENCE

Embedded Systems Software Engineer Intern

Jan 2027

Tesla AI

Palo Alto, California, USA

- Incoming on AI Platforms team, supporting Optimus & Tesla AP in January 2027

Software Development Engineer Intern

May 2026 – Present

Amazon

Vancouver, BC, Canada

- Built the **agent harness** for an LLM ticket-triage platform on AWS Bedrock (tool integration via **MCP**, structured output parsing, fallback chains), delivering root-cause analysis for **2,000+ production tickets** across 8 Amazon teams
- Engineered **context and retrieval pipelines** grounding every analysis in team runbooks and **760+ ingested ticket resolutions** (2,800+ RAG retrievals), with source citations on agent responses
- Designed **guardrails for autonomous actions** (confidence gates, default-deny allowlists), enabling the agent to safely post **1,900+ ticket responses** and resolve **90+ tickets end-to-end** without human intervention

Embedded Software Engineer Co-op

Jan 2025 – May 2026

General Dynamics

Calgary, AB, Canada / Remote

- **Saved \$20k+ per vehicle platform** by enabling an additional user on existing hardware, eliminating the need for another communications unit
- Developed **3 Linux device drivers and daemons in C/C++** for tactical communications systems spanning **20+ networks** of 4-8 distributed nodes
- Implemented **configurable layer 3/4/5 networking** for communication across distributed embedded systems
- **Reduced CI/CD pipeline runtime by 50%** using build caching and dedicated Docker build environments to eliminate redundant rebuilds
- Built lab-local **AI infrastructure and agents** for testing and development, supporting **20+ engineers** and 3 build pipelines

EDUCATION

The University of British Columbia

Vancouver, BC, Canada

BSc Computer Science & Statistics (GPA 4.0/4.33)

Sep 2022 – Dec 2027

Relevant Coursework: Operating Systems, Machine Learning & Data Mining, Computer Vision, Algorithm Design & Analysis, Statistical Learning, Applied Linear Algebra, Computer Graphics, Stochastic Processes

PROJECTS

FITNET - Real-Time Wearable Motion Tracking | *C, Swift, iOS, FreeRTOS, BLE*

Feb – Dec 2025

- Built a real-time wearable motion-tracking system using **ESP32-S3 nodes, EMG, and 9-axis IMUs**, with sub-millisecond synchronization across distributed sensors
- Achieved **95% movement-classification accuracy** using DTW-KNN on synchronized EMG and motion data, with on-device orientation estimation using a Mahony AHRS
- Streamed and processed sensor data over optimized **Bluetooth Low Energy** in a SwiftUI/SceneKit iOS app for real-time **3D motion visualization**

Find My Force - RF Signal Classification | *Python, PyTorch, XGBoost, scikit-learn*

Mar 2026

- Built an **RF signal classification pipeline** combining CNN-extracted features with statistical signal features for transmitter identification
- Achieved **89% classification accuracy** by ensembling PyTorch CNN, XGBoost, and logistic regression models

TECHNICAL SKILLS

Languages: Python, C++, C, Java, C#, TypeScript, JavaScript, Swift, Rust, Kotlin, R, GLSL

Frameworks & Libraries: PyTorch, scikit-learn, XGBoost, Three.js, Node.js, .NET, SwiftUI, OpenGL, FreeRTOS

Cloud & AI: AWS, AWS CDK, AWS Bedrock, Firebase, MCP, RAG, Supervised & Unsupervised Learning

Systems & Developer Tools: Linux, Git, Docker, Jenkins, CMake, Maven, gdb, Bash